When Does Aggregating Explanations Work?
OrganCMNIST Results

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This report consolidates the released `OrganCMNIST` experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Result source snapshot: `d6e4b9bbfb3c` from Workshop-When-Does-Explanations.

Table 1: NAIVE results for `OrganCMNIST` with `ViT-B/16` (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 8216$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	0.039	0.357	0.104	0.421	0.047	-0.174	0.040	-0.244	0.029	-0.176	0.015	-0.205	0.010	-0.011	-0.004	-0.040	0.071	-0.032	0.079	-0.029
Simple Averaging	0.023	0.369	0.082	0.434	0.033	-0.176	0.048	-0.243	0.021	-0.172	0.024	-0.199	0.004	-0.015	0.006	-0.041	0.023	-0.084	0.026	-0.083
Borda	0.021	0.379	0.081	0.442	0.038	-0.169	0.048	-0.236	0.021	-0.167	0.024	-0.194	0.007	-0.017	0.007	-0.044	0.026	-0.051	0.030	-0.049
Kemeny–Young	0.021	0.381	0.080	0.444	0.041	-0.163	0.054	-0.231	0.021	-0.157	0.028	-0.185	0.005	-0.011	0.005	-0.038	0.023	-0.061	0.026	-0.061
RRF	0.022	0.375	0.082	0.439	0.039	-0.165	0.051	-0.232	0.024	-0.154	0.031	-0.181	0.005	-0.006	0.004	-0.033	0.021	-0.063	0.023	-0.063
Schulze	0.020	0.380	0.080	0.443	0.041	-0.161	0.053	-0.228	0.025	-0.162	0.028	-0.191	0.005	-0.008	0.006	-0.036	0.024	-0.056	0.027	-0.057

Table 2: IND results for `OrganCMNIST` with `ViT-B/16` (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 8216$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	baseline	0.018	0.408	0.054	0.478	0.004	-0.214	0.032	-0.277	0.002	-0.222	0.014	-0.241	-0.004	-0.060	0.003	-0.077	0.041	-0.034	0.048	-0.044
Simple Averaging	matched NAIVE	0.011	0.433	0.042	0.504	0.004	-0.213	0.030	-0.275	0.003	-0.215	0.017	-0.238	-0.006	-0.056	0.005	-0.082	0.035	-0.038	0.041	-0.039
Simple Averaging	IND	0.009	0.466	0.041	0.539	0.005	-0.183	0.033	-0.241	0.005	-0.194	0.019	-0.215	-0.001	-0.083	0.011	-0.107	0.040	-0.045	0.044	-0.045
Borda	matched NAIVE	0.008	0.413	0.046	0.484	0.009	-0.219	0.036	-0.279	0.008	-0.221	0.020	-0.241	0.001	-0.061	0.009	-0.084	0.038	-0.042	0.040	-0.044
Borda	IND	0.012	0.424	0.050	0.495	0.010	-0.213	0.037	-0.274	0.006	-0.220	0.016	-0.242	0.000	-0.087	0.009	-0.111	0.040	-0.046	0.041	-0.048
Kemeny–Young	matched NAIVE	0.010	0.429	0.047	0.502	0.013	-0.200	0.041	-0.259	0.008	-0.205	0.023	-0.223	0.001	-0.047	0.011	-0.068	0.039	-0.040	0.042	-0.040
Kemeny–Young	IND	0.014	0.435	0.052	0.507	0.011	-0.199	0.040	-0.259	0.008	-0.212	0.022	-0.231	0.000	-0.061	0.013	-0.084	0.038	-0.042	0.041	-0.043
RRF	matched NAIVE	0.010	0.422	0.048	0.494	0.011	-0.211	0.040	-0.270	0.007	-0.210	0.020	-0.231	-0.002	-0.049	0.009	-0.072	0.036	-0.042	0.039	-0.044
RRF	IND	0.011	0.433	0.051	0.503	0.015	-0.201	0.041	-0.262	0.008	-0.210	0.019	-0.231	0.000	-0.057	0.011	-0.081	0.042	-0.041	0.043	-0.045
Schulze	matched NAIVE	0.009	0.421	0.047	0.493	0.012	-0.208	0.040	-0.267	0.008	-0.213	0.021	-0.232	-0.001	-0.053	0.008	-0.074	0.037	-0.040	0.041	-0.041
Schulze	IND	0.013	0.428	0.051	0.499	0.011	-0.205	0.039	-0.265	0.005	-0.220	0.018	-0.239	0.000	-0.075	0.012	-0.099	0.041	-0.043	0.043	-0.045

Table 3: NOISE-S results for `OrganCMNIST` with `ViT-B/16` (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.033	0.357	0.096	0.419	0.037	-0.181	0.043	-0.252	0.024	-0.175	0.018	-0.205	0.001	-0.014	-0.002	-0.042	0.029	-0.074	0.029	-0.075
Borda	3	0.035	0.346	0.099	0.408	0.039	-0.187	0.038	-0.259	0.022	-0.189	0.014	-0.218	0.003	-0.020	-0.004	-0.050	0.063	-0.038	0.071	-0.038
Kemeny–Young	3	0.042	0.350	0.107	0.415	0.048	-0.181	0.043	-0.250	0.025	-0.179	0.015	-0.206	0.003	-0.012	-0.007	-0.041	0.063	-0.036	0.070	-0.033
RRF	3	0.041	0.346	0.104	0.409	0.041	-0.181	0.041	-0.254	0.024	-0.183	0.018	-0.212	0.001	-0.007	-0.004	-0.039	0.059	-0.039	0.066	-0.038
Schulze	3	0.038	0.344	0.103	0.407	0.037	-0.190	0.038	-0.261	0.021	-0.189	0.013	-0.217	0.002	-0.021	-0.003	-0.051	0.064	-0.037	0.071	-0.037

Table 4: NOISE-K results for `OrganCMNIST` with `ViT-B/16` (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.033	0.357	0.096	0.419	0.037	-0.181	0.043	-0.252	0.024	-0.175	0.018	-0.205	0.001	-0.014	-0.002	-0.042	0.029	-0.074	0.029	-0.075
Borda	3	0.035	0.346	0.099	0.408	0.039	-0.187	0.038	-0.259	0.022	-0.189	0.014	-0.218	0.003	-0.020	-0.004	-0.050	0.063	-0.038	0.071	-0.038
Kemeny–Young	3	0.042	0.350	0.107	0.415	0.048	-0.181	0.043	-0.250	0.025	-0.179	0.015	-0.206	0.003	-0.012	-0.007	-0.041	0.063	-0.036	0.070	-0.033
RRF	3	0.041	0.346	0.104	0.409	0.041	-0.181	0.041	-0.254	0.024	-0.183	0.018	-0.212	0.001	-0.007	-0.004	-0.039	0.059	-0.039	0.066	-0.038
Schulze	3	0.038	0.344	0.103	0.407	0.037	-0.190	0.038	-0.261	0.021	-0.189	0.013	-0.217	0.002	-0.021	-0.003	-0.051	0.064	-0.037	0.071	-0.037